

Package ‘JEL’

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Title Jointly Estimated Landmarking for Dynamic Prediction

Version 2.2

Description Implements Jointly Estimated Landmarking (JEL) methods for dynamic prediction using longitudinal and survival data. Supports models with time-varying coefficients and multiple longitudinal processes with a fast EM algorithm. For detailed methodology, refer to Jinwoo Cho and Kehui Chen, "Dynamic Prediction of Survival Outcomes Using Jointly Estimated Landmarking Approach."

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Description

This function computes the dynamic area under the curve (AUC) for survival predictions at a specified landmark time and prediction horizon.

Usage

```
AUCdyn(
  pred_surv_result,
  data,
  landmarks,
  tau,
  var_list = list(id = "id", EvTime = "EvTime", event = "event")
)
```

Arguments

<code>pred_surv_result</code>	The result from the <code>predict.JEL</code> function. It can be a <code>data.frame</code> or a <code>list</code> : - If a <code>data.frame</code> , it should include columns for <code>id</code> , <code>tau</code> , and survival probabilities. - If a <code>list</code> , it should include multiple survival prediction outputs.
<code>data</code>	A <code>data.frame</code> containing the survival data, including subject IDs, event times, and event statuses.
<code>landmarks</code>	A numeric value specifying the landmark time for dynamic prediction.
<code>tau</code>	A numeric value specifying the prediction horizon.
<code>var_list</code>	A <code>list</code> specifying the column names in the data: - <code>id</code> : Column name for the subject IDs (default: <code>"id"</code>). - <code>EvTime</code> : Column name for the event times (default: <code>"EvTime"</code>). - <code>event</code> : Column name for the event status (default: <code>"event"</code>).

Details

The function calculates the dynamic AUC by comparing predicted survival probabilities from `predict.JEL` against actual survival outcomes at the specified landmark and prediction horizon. It uses Kaplan-Meier estimates to adjust survival probabilities for specific cases.

Value

A numeric value representing the dynamic AUC.

Description

This function constructs confidence bands for the time-varying coefficients of a Jointly Estimated Landmarking (JEL) model. It requires a B-spline configuration (Bs) to compute the confidence intervals for both slope and intercept terms.

Usage

```
confBands.JEL(JEL_models, K = 1, alpha = 0.05)
```

Arguments

JEL_models	A fitted JEL model object of class "JEL", containing the estimated joint model components.
K	A numeric value indicating the index of the longitudinal process for which confidence bands are computed. - Default is 1, which applies to single-process models.
alpha	A numeric value specifying the significance level for the confidence bands. Default is 0.05 (95% confidence level).

Details

The confBands.JEL function computes confidence bands for the time-varying slope and intercept coefficients in a JEL model using the following steps:

1. B-Spline Configuration:

- Extracts the B-spline configuration (Bs) from the fitted JEL model to construct the design matrix for the time variable.
- Ensures the spline configuration is properly indexed for multi-process models.

2. Variance-Covariance Matrix Adjustment:

- Performs eigen decomposition of the variance-covariance matrix to ensure it is positive semi-definite.
- Sets any negative eigenvalues to zero before recomputing the matrix.

3. Confidence Band Computation:

- Computes point estimates for the time-varying coefficients (slope and intercept) based on the B-spline design matrix.
- Uses the adjusted variance-covariance matrix to calculate the standard errors.
- Constructs confidence intervals for each coefficient using the chi-square distribution.
- Confidence bands are calculated based on the chi-square distribution, with critical values determined by the specified significance level (alpha) and the degrees of freedom (df) derived from the B-spline configuration.

Value

A list containing two data frames:

- intercept: A data frame with columns tau (prediction time after the landmarking time point), est (point estimate), lower (lower confidence limit), and upper (upper confidence limit) for the intercept term.
- slope: A data frame with the same structure as intercept, providing the confidence bands for the slope term.

JEL1	<i>Jointly Estimated Landmarking (JEL) for a Singular Longitudinal Process</i>
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Description

The JEL1 function implements a Jointly Estimated Landmarking approach to predict survival probabilities at specified landmark times. It integrates longitudinal mixed-effects modeling and Cox proportional hazards modeling, providing dynamic predictions for survival outcomes based on past longitudinal data.

Usage

```
JEL1(
  data,
  land_time,
  u,
  lme_fixed,
  lme_random,
  cox_formula,
  Bs = NULL,
  rho = 0,
  time_var
)
```

Arguments

- | | |
|-----------|---|
| data | A data frame containing both longitudinal and survival data. The data must include the following columns: id, time variable for the longitudinal data, survival event time, and survival event indicator (censor). All these variables must be referenced in lme_fixed, lme_random, and cox_formula. Refer to pbc2 as an example. |
| land_time | A numeric value specifying the landmarking time point for dynamic survival prediction. |
| u | A numeric value specifying the interval length of past observations to include in the modeling. |
| lme_fixed | <p>A list containing fixed-effects formulas for the longitudinal mixed-effects models.</p> <p>- Must contain exactly one formula of fixed effect of linear mixed model.</p> |

lme_random	A list containing random-effects formulas for the longitudinal mixed-effects models. - Must contain exactly one formula of the form <code>~ time id</code> .
cox_formula	A formula for the Cox proportional hazards model, specifying the survival outcome and covariates. - Must follow the format <code>Surv(time, event) ~ covariates</code> .
Bs	An optional list specifying B-spline parameters for time-varying effects, with components: - df: Degrees of freedom. - degree: Degree of the spline. - knots: Locations of internal knots. - Bknots: Boundary knots. If NULL, the landmarking model is fit without time-varying effects (default: NULL).
rho	A numeric value specifying the penalty parameter for the landmarking model (default: 0).

Details

The Jointly Estimated Landmarking (JEL) framework dynamically combines longitudinal and survival data to estimate survival probabilities at specified landmark times. It involves preparing data, fitting longitudinal mixed-effects models and Cox proportional hazards models, and integrating these into a unified framework.

Value

A list of class "JEL" containing:

fitLME_list	A list of fitted longitudinal mixed-effects models, with estimated fixed and random effects.
fitCOX	The fitted Cox proportional hazards model, including coefficients, baseline hazard estimates, and diagnostics.
fitJEL	The fitted landmarking model that combines results from the longitudinal and survival models, with sub-items:
coefficients	A list containing: - beta Estimated fixed effects for the longitudinal process. - phi Estimated baseline survival covariates. - eta Estimated association parameters between longitudinal and survival processes. - Ysigma Residual variance of the longitudinal process. - Bsigma Random effects variance-covariance matrix. - lamb Estimated baseline hazard function over time.
logLik	Log-likelihood of the fitted landmarking model.
numIter	Number of EM algorithm iterations performed.
Vcov	Variance-covariance matrix of the parameter estimates.
est.bi	Predicted random effects for each subject.
convergence	Indicator of convergence status ("success" or "failure").
control	Control parameters used for model fitting, including tolerance and maximum iterations.

time.SE Computation time for standard error estimation.
 N Total number of observations in the longitudinal dataset.
 n Number of unique subjects in the dataset.
 d Event indicator vector for survival data (1 if the event occurred, 0 otherwise).
 rho Penalty parameter used in the landmarking model.
 dataMat A list summarizing the prepared data, including:
 B Basis matrix for time-varying effects (if applicable).
 ID Subject ID vector.

dataset A list of prepared datasets used in the JEL model:

LMM_dat The dataset formatted for the longitudinal mixed-effects models. And the data is transformed the time variable given s and u.
 Surv_dat The dataset formatted for the Cox proportional hazards model. And the data involves start, stop, and event variable for JEL model.

Examples

```
# Load example dataset
data("pbc2")
str(pbc2)

# Specify landmarking parameters
s <- 5
u <- 3
tau <- 2

# Example 1: JEL with rho = 0 (Cox proportional hazards model)
testJEL1 <- JEL1(
  pbc2, land_time = s, u = u,
  lme_fixed = list(log(serBilir) ~ year),
  lme_random = list(~ year | id),
  cox_formula = Surv(years, status2) ~ drug,
  Bs = NULL,
  rho = 0
)

# Example 2: JEL with rho = 1 (Proportional odds model)
testJEL2 <- JEL1(
  pbc2, land_time = s, u = u,
  lme_fixed = list(log(serBilir) ~ year),
  lme_random = list(~ year | id),
  cox_formula = Surv(years, status2) ~ drug,
  Bs = NULL,
  rho = 1
)

# Example 3: JEL using time-varying effects with B-splines
testJEL3 <- JEL1(
  pbc2, land_time = s, u = u,
  lme_fixed = list(log(serBilir) ~ year),
  lme_random = list(~ year | id),
  cox_formula = Surv(years, status2) ~ drug,
  Bs = list(df = 3, degree = 1, knots = NULL, Bknots = NULL),
  rho = 0
)
```

```

# Predictions of conditional survival probabilities,  $\Pr(T > \tau + s \mid T > s)$ 
prediction_result1 <- predict(testJEL1, tau = tau)
prediction_result2 <- predict(testJEL2)
prediction_result3 <- predict(testJEL3, tau = tau)

# Confidence bands for  $\eta_0(t)$  and  $\eta_1(t)$  from time-varying model
conint_tv <- confBands.JEL(testJEL3, alpha = 0.05)

# AUC calculations
AUCtest1 <- AUCdyn(
  pred_surv_result = prediction_result1,
  data = pbc2, landmarks = s, tau = tau,
  var_list = list(id = "id", EvTime = "years", event = "status2")
)

# PE calculations
PEtest1 <- PEDyn(
  pred_surv_result = prediction_result1,
  data = pbc2, landmarks = s, tau = tau,
  var_list = list(id = "id", EvTime = "years", event = "status2")
)

```

JEL_dat	<i>Create landmarked longitudinal and survival datasets for JEL/LLAJEL</i>
---------	--

Description

Splits an individual-level longitudinal dataset into:

- LMM_dat: the full longitudinal record for each retained subject (all observations, including post-landmark rows). Subjects are retained if they are at risk at the landmark ($\text{EvTime} > s$) and have at least one longitudinal observation at or before s (so a landmark baseline can be defined); and
- Surv_dat: subject-level survival dataset starting at the landmark time, with $\text{start} = 0$, $\text{stop} = \text{EvTime} - s$, and event indicator carried over.

Usage

```

JEL_dat(
  data,
  land_time,
  negative = FALSE,
  var_list = list(id = "id", time = "time", EvTime = "EvTime", event = "event"),
  h = NULL,
  min_window_obs = NULL,
  y_vars = NULL
)

```

Arguments

<code>data</code>	A data frame containing longitudinal rows with at least the columns in <code>var_list</code> (<code>id</code> , <code>time</code> , <code>EvTime</code> , <code>event</code>).
<code>land_time</code>	Numeric scalar landmark time s . Subjects are kept only if <code>EvTime</code> > s .
<code>negative</code>	Logical. If TRUE, time is shifted to <code>time</code> - s and the longitudinal window is $[-s, 0]$. If FALSE (default), time is kept on the original scale and the window is $[0, s]$.
<code>var_list</code>	A named list specifying column names in <code>data</code> . Defaults to <code>list(id = "id", time = "time", EvTime = "EvTime", event = "event")</code> .
<code>h</code>	Numeric scalar bandwidth (half-window width) used together with <code>min_window_obs</code> to define the complete-case filter. Only consulted when <code>min_window_obs</code> is non-NULL. Default NULL.
<code>min_window_obs</code>	Optional integer. If non-NULL, applies a complete-case filter: a subject is retained only if it has at least <code>min_window_obs</code> non-missing observations of <i>every</i> marker in <code>y_vars</code> inside the kernel window $ t - s \leq h$ (for <code>negative</code> = TRUE the window is centred at 0). Default NULL (no filtering; all at-risk subjects with an observation up to s are kept).
<code>y_vars</code>	Character vector of longitudinal outcome column names checked by the <code>min_window_obs</code> complete-case filter. Required when <code>min_window_obs</code> is non-NULL. Default NULL.

Details

The function also attaches `start/stop/event` back to `LMM_dat` (one row per longitudinal observation) to match older JEL-style workflows.

Steps performed:

1. Coerces `id` column to numeric.
2. Keeps only subjects at risk at the landmark: `EvTime` > s .
3. Keeps only subjects with at least one longitudinal observation at or before s .
4. `LMM_dat` retains ALL longitudinal rows for those subjects (no $t \leq s$ cutoff).
5. Builds `Surv_dat` with `start` = 0, `stop` = `EvTime` - s , and `event` from the subject-level record. Baseline covariates at the landmark are taken as the last observed row in the longitudinal window (`dplyr::last`).
6. Left-joins `start/stop/event` back onto `LMM_dat`.

Value

A list with components:

- `LMM_dat`: full longitudinal data for retained subjects, with `start/stop/event` attached.
- `Surv_dat`: subject-level survival data with `start/stop/event` and baseline covariates.
- `s`: the landmark time used.
- `var_list`: the column mapping used.

Examples

```
## Not run:
dat_split <- JEL_dat(
  data = dat,
  land_time = 2,
  negative = FALSE,
  var_list = list(id="id", time="time", EvTime="Time", event="event")
)
head(dat_split$LMM_dat)
head(dat_split$Surv_dat)

## End(Not run)
```

LLAJEL

Fit an LLA-based Jointly Estimated Landmarking (LLAJEL) model

Description

Fits a landmarked joint model using Local Linear Approximation (LLA) for the longitudinal component and a Cox model for the survival component, following a JEL2-style workflow: 'prep -> fit -> return' a '"JEL"' object.

Usage

```
LLAJEL(
  train_dataset,
  y_vars,
  s,
  h,
  base_terms,
  ker = "gaussian",
  model = "Fixed",
  Bs = NULL,
  gh.nodes = 5,
  max.iter = 200,
  tol = 0.01,
  diff.type = "abs.rel",
  collect.hist = TRUE,
  Vcov = TRUE,
  verbose = FALSE,
  update_c = TRUE,
  h_init = NULL
)
```

Arguments

<code>train_dataset</code>	A list containing at least <code>LMM_dat</code> , <code>Surv_dat</code> , and <code>var_list</code> . <code>Surv_dat</code> must already be in (start, stop, event) form appropriate for <code>Surv(start, stop, event)</code> .
<code>y_vars</code>	Character vector of longitudinal outcome names to be modeled by LLA.
<code>s</code>	Landmark time (scalar).

<code>h</code>	Bandwidth for kernel weighting in the LLA landmarking step (scalar).
<code>base_terms</code>	Character vector (or formula components) for baseline Cox terms to include in the survival submodel alongside LLA-derived terms.
<code>ker</code>	Kernel type for landmark weighting. Default is "gaussian".
<code>model</code>	Longitudinal model type: "Fixed" for constant landmark associations or "TimeVar" for time-varying (B-spline) associations. Default is "Fixed".
<code>Bs</code>	Optional list of B-spline specifications (one per longitudinal outcome) used only when <code>model = "TimeVar"</code> ; each element is a list with entries <code>df</code> , <code>degree</code> , <code>knots</code> , <code>Bknots</code> (or <code>NULL</code> to leave a marker time-invariant). Default is <code>NULL</code> (all fixed).
<code>gh.nodes</code>	Number of Gauss-Hermite nodes used by the EM step. Default is 5.
<code>max.iter</code>	Maximum number of EM iterations. Default is 200.
<code>tol</code>	Convergence tolerance. Default is 0.01.
<code>diff.type</code>	Convergence criterion type passed to the EM routine. Default is "abs.rel".
<code>collect.hist</code>	Logical; whether to collect iteration history. Default is <code>TRUE</code> .
<code>Vcov</code>	Logical; whether to compute/post-process variance-covariance. Default is <code>TRUE</code> .
<code>verbose</code>	Logical; verbosity for the EM routine. Default is <code>FALSE</code> .
<code>update_c</code>	Logical; if <code>TRUE</code> (default) the population mean of the random effects <code>c</code> is re-estimated at each EM iteration as the M-step update $\hat{c} = n_s^{-1} \sum_i \hat{b}_i$. If <code>FALSE</code> , <code>c</code> is held fixed at its initial value (legacy behaviour).
<code>h_init</code>	Optional bandwidth used only to construct the initial local-linear BLUPs of the random effects, decoupled from the fitting bandwidth <code>h</code> . If <code>NULL</code> (default) the fitting bandwidth <code>h</code> is reused for initialization.

Details

This function assumes the following helpers exist in the package namespace:

- `prep_LLA_landmark()`
- `make_cox_formula_from_LLA()`
- `InitVal_LLAJEL()`
- `RefinedfastEM_LLA()`

Internally, this function:

1. Builds landmarked LLA objects via `prep_LLA_landmark()`.
2. Constructs and fits a Cox model on `prep$urv_s2` using `make_cox_formula_from_LLA()` and `survival::coxph()`.
3. Builds indexing/design objects for the EM step.
4. Initializes survival parameters via `InitVal_LLAJEL()` and runs the EM algorithm via `RefinedfastEM_LLA()`.
5. Returns an object of class "JEL" with estimates and optional diagnostics.

Value

An object of class "JEL" (a list) containing at least:

- `coefficients`: estimated fixed effects / parameters (from EM output).
- `Vcov`: estimated variance-covariance matrix (if computed).
- `est.bi`: estimated random effects (if returned by EM).

- convergence: "success" or "failure" based on iteration history.
- history: EM iteration history (if collected).
- n, d: subject count and event indicators.
- fitCOX: fitted Cox model object (for debugging/inspection).
- prep: preprocessing output (for debugging/inspection).
- dataset: echoes key inputs and settings.

Examples

```
## Not run:
fit <- LLAJEL(
  train_dataset = train_dataset,
  y_vars = c("Y.1", "Y.2", "Y.3"),
  s = 2, h = 2,
  base_terms = c("age", "sex")
)
print(fit$coefficients)

## End(Not run)
```

PEdyn

Dynamic Prediction Error (PE) Calculation

Description

This function computes the dynamic prediction error (PE) for survival predictions at a specified landmark time and prediction horizon.

Usage

```
PEdyn(
  pred_surv_result,
  data,
  landmarks,
  tau,
  var_list = list(id = "id", EvTime = "EvTime", event = "event")
)
```

Arguments

pred_surv_result	The result from the predict.JEL function. It can be a data.frame or a list: - If a data.frame, it should include columns for id, tau, and survival probabilities. - If a list, it should include multiple survival prediction outputs.
data	A data.frame containing the survival data, including subject IDs, event times, and event statuses.
landmarks	A numeric value specifying the landmark time for dynamic prediction.
tau	A numeric value specifying the prediction horizon.

`var_list` A list specifying the column names in the data:

- `id`: Column name for the subject IDs (default: "id").
- `EvTime`: Column name for the event times (default: "EvTime").
- `event`: Column name for the event status (default: "event").

Details

The function calculates the dynamic PE by comparing the predicted survival probabilities from `predict.JEL` against actual survival outcomes at a specified landmark time and prediction horizon. The prediction error is computed using Kaplan-Meier estimates to adjust for censored cases.

Value

A numeric value representing the dynamic PE.

<code>predict.JEL</code>	<i>Predict Conditional Survival Probabilities Using JEL Models</i>
--------------------------	--

Description

This function predicts conditional survival probabilities using JEL (Joint Estimation with Landmarks) models.

Usage

```
## S3 method for class 'JEL'
predict(
  fitLLAJEL,
  testdat = NULL,
  tau = NULL,
  CI = FALSE,
  MC = 100,
  alpha = 0.05,
  rho = 0,
  bLLA_new = NULL
)
```

Arguments

<code>testdat</code>	An optional data frame containing new data for prediction. If <code>NULL</code> , the training dataset from <code>JEL_models</code> is used (default: <code>NULL</code>).
<code>tau</code>	A numeric value specifying the prediction time horizon. If <code>NULL</code> , the full conditional survival probabilities are returned (default: <code>NULL</code>).
<code>CI</code>	A logical value indicating whether to compute confidence intervals for the predicted survival probabilities (default: <code>FALSE</code>). The confidence interval is obtained using Monte Carlo approach.
<code>MC</code>	An integer specifying the number of Monte Carlo simulations for confidence interval estimation (default: 100).
<code>alpha</code>	A numeric value specifying the significance level for confidence interval estimation (default: 0.05).

JEL_models A list from the JEL1 or JEL2, including:

- fitJEL: The fitted Joint Estimation Landmarking (JEL) model.
- fitLME_list: A list of fitted longitudinal linear mixed-effects (LME) models.
- fitCOX: The fitted Cox proportional hazards model.
- dataset: A list of training and testing datasets, with the following components:
 - LMM_dat: Data used for the longitudinal mixed-effects models.
 - Surv_dat: Data used for the survival Cox model.
- s: Landmark time used for model training.
- u: Interval length of past observations included in the modeling.
- var_list: A list of variable names for identification (id), time (time), survival event time (EvTime), and event indicator (event).

Details

This function computes conditional survival probabilities for individuals using the components of a JEL model. It estimates the Best Linear Unbiased Predictors (BLUPs) for random effects based on the new or training dataset and combines them with survival predictions from the fitted Cox model. If a prediction time horizon ('tau') is specified, the function filters the results to include survival probabilities at or before the specified time horizon.

The workflow involves:

1. BLUP Estimation: Predicts the random effects for new data using the fitted JEL and LME models.
2. Survival Probability Computation: Uses the Cox model and the estimated BLUPs to calculate conditional survival probabilities.
3. Filtering by tau (if provided): Returns survival probabilities up to the specified time horizon.

Value

A data frame or list containing the predicted survival probabilities. The structure of the output depends on whether tau is specified:

- If tau is NULL, the function returns a full prediction for all time points.
- If tau is specified, the function returns predictions filtered at or before the given time horizon.

select_h_longitudinal *Longitudinal LSCV bandwidth selection for LLAJEL*

Description

Selects the kernel bandwidth h for the Local Linear Approximation (LLA) step of LLAJEL by leave-one-observation-out cross-validation on the longitudinal data, evaluated via the PRESS leverage identity (no obs-by-obs refit required).

Usage

```
select_h_longitudinal(
  LMM_dat,
  Surv_dat,
  y_vars,
  s0,
  h_grid,
  ker = c("epanechnikov", "gaussian"),
  W = NULL,
  var_list = list(id = "id", time = "time"),
  baseline_h = NULL,
  fit = NULL,
  train_dataset = NULL,
  base_terms = character(0),
  Bs = NULL,
  gh.nodes = 7,
  max.iter = 1000,
  tol = 0.001,
  verbose = FALSE
)
```

Arguments

LMM_dat	Long-format longitudinal data frame containing the columns named in <code>var_list</code> plus the outcome columns listed in <code>y_vars</code> .
Surv_dat	Survival data frame (subject-level) containing the subject id column named in <code>var_list</code> . Used to enumerate the subject roster.
y_vars	Character vector of longitudinal outcome column names.
s0	Landmark time (scalar).
h_grid	Numeric vector of candidate bandwidths (length ≥ 2 , all > 0).
ker	Kernel: one of "epanechnikov" (default) or "gaussian".
W	Deprecated and ignored. The LOO-CV criterion is now global (uses all observations); this argument is kept only so existing callers that still pass <code>W = ...</code> do not error.
var_list	Named list with at least <code>id</code> and <code>time</code> pointing to the corresponding column names in <code>LMM_dat</code> / <code>Surv_dat</code> .
baseline_h	Bandwidth used for the one-shot baseline EM fit (<code>LLAJEL</code>). Defaults to <code>median(h_grid)</code> .
fit	Optional pre-fitted "JEL" object (the output of <code>LLAJEL</code>) at <code>baseline_h</code> . If supplied, the baseline EM is skipped and <code>D</code> , <code>var.e</code> are taken from this object.
train_dataset	Training-dataset list of the form expected by <code>LLAJEL</code> . Required when <code>fit</code> is <code>NULL</code> .
base_terms, Bs, gh.nodes, max.iter, tol	Passed through to <code>LLAJEL</code> when running the baseline EM.
verbose	Logical; if <code>TRUE</code> , prints per-h CV values during the sweep.

Details

The criterion is **GLOBAL**: every longitudinal observation enters the LOO-CV (there is no window around the landmark), so candidate bandwidths are compared on the full observation set. Population

covariance D and noise variance var.e are taken from a single baseline LLAJEL fit at `baseline_h` (default `median(h_grid)`); the centering c (cLLA) is refreshed at every candidate h . For each subject i and candidate h , the BLUP precision is

$$\Lambda_i = D^{-1} + Z_i^\top R_i^{-1} Z_i, \quad R_i = \text{diag}(\sigma_k^2/w_{ij}),$$

the posterior mean is

$$\mu_i = c + \Lambda_i^{-1} Z_i^\top R_i^{-1} (Y_i - Z_i c),$$

and the leave-one-out residual at obs (j, k) follows from

$$e_{LOO,jk} = \frac{Y_{ijk} - z_{jk}^\top \mu_i}{1 - h_{jk}}, \quad h_{jk} = \frac{w_{ij}}{\sigma_k^2} z_{jk}^\top \Lambda_i^{-1} z_{jk}.$$

For obs outside the kernel support at this h (i.e. $w_{ij} = 0$), $h_{jk} = 0$ and the LOO residual reduces to the regular BLUP residual.

Per-process selection minimises $\text{MSE}_k(h)$ separately; the shared criterion is $\sum_k \text{MSE}_k(h)/\text{var.e}_k$, normalised by baseline-fit noise scale so processes with different magnitudes are comparable.

Value

A list with components:

`h_grid` The input candidate grid.

`cv_per_process` Matrix $|h_grid| \times nK$ of per-process LOO MSEs.

`cv_shared` Numeric vector of scale-normalised shared criterion values.

`h_per_process` Length- nK vector of per-process minimisers.

`h_shared` Single minimiser of `cv_shared`.

`n_in_W_per_proc` Per-process obs counts inside the criterion window (identical across rows by construction).

`W` Always Inf (criterion is global; window removed).

`ker` Kernel used.

`baseline_h` Bandwidth at which the baseline EM was run.

`baseline_fit` The baseline "JEL" fit object.

See Also

[LLAJEL](#)

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